

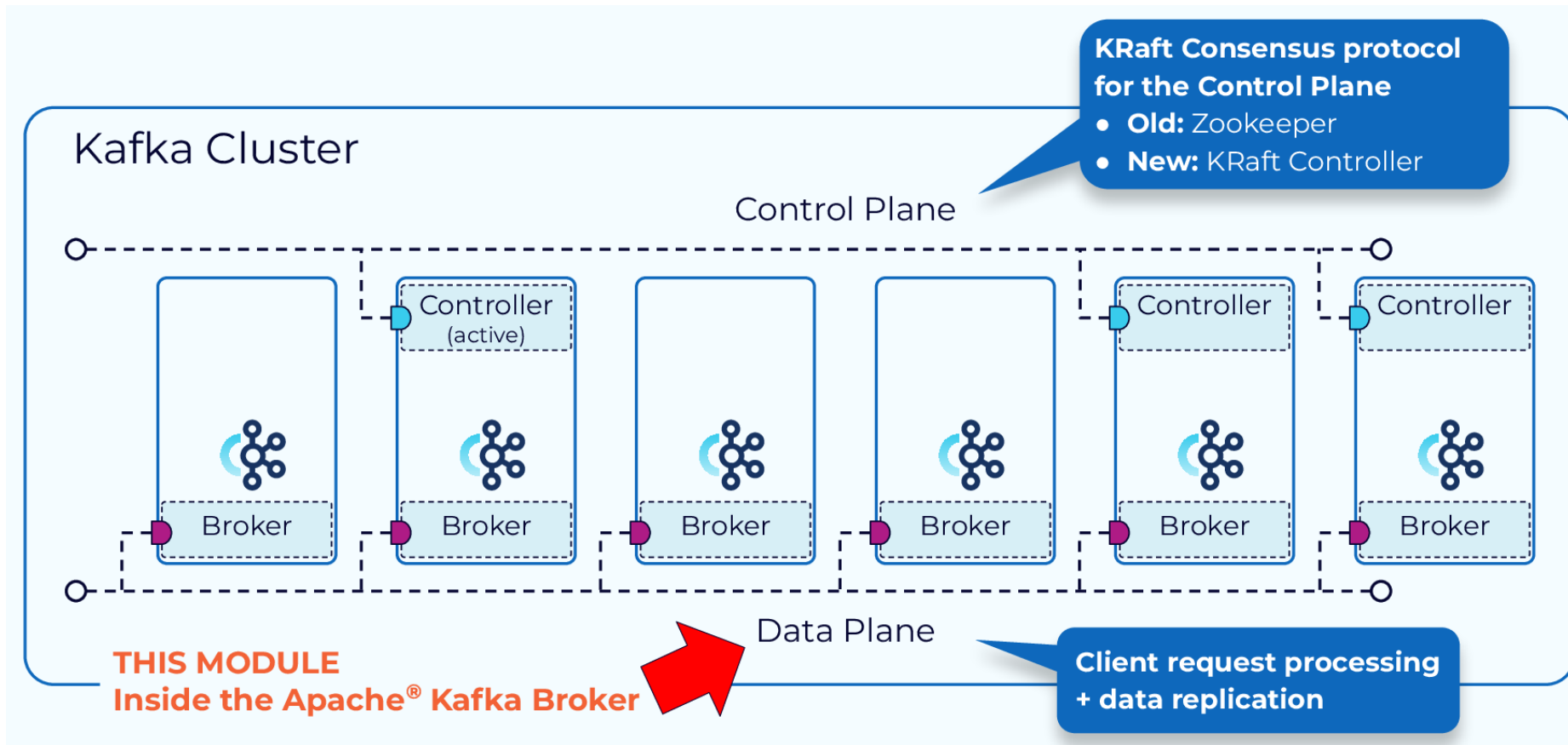
Control Pane - KRaft

KRaft mode (Kafka Raft) is now generally available, removing the need for **ZooKeeper** in Kafka's architecture. KRaft is a consensus protocol that consolidates metadata management into Kafka itself, simplifying the system by eliminating the separate ZooKeeper component.

Key Points:

1. **Simplification:** KRaft reduces complexity by eliminating ZooKeeper, a separate system with its own configuration and management tools. This makes deploying and managing Kafka clusters easier.
2. **Benefits of KRaft:**
 - Enables **right-sized clusters** that scale efficiently to millions of partitions.
 - Provides **near-instantaneous metadata failover** for higher availability.
 - Creates a **single management model** for authentication, configuration, and protocols, making administration easier.
3. **Controller Quorum:** KRaft uses a **Raft quorum** of controller nodes to manage Kafka's metadata log. The active controller handles all metadata updates, while follower controllers replicate the log and provide failover support. This results in more efficient metadata propagation and faster failover recovery.
4. **Scalability:** KRaft supports a significantly larger number of partitions compared to ZooKeeper-based Kafka. While ZooKeeper limited clusters to about 200,000 partitions, KRaft can support up to 2 million partitions, with improved shutdown and recovery times.
5. **Fault Tolerance:** Like ZooKeeper, KRaft requires a majority of controller nodes to be operational for continued functionality. For example, a three-node cluster can survive one failure.

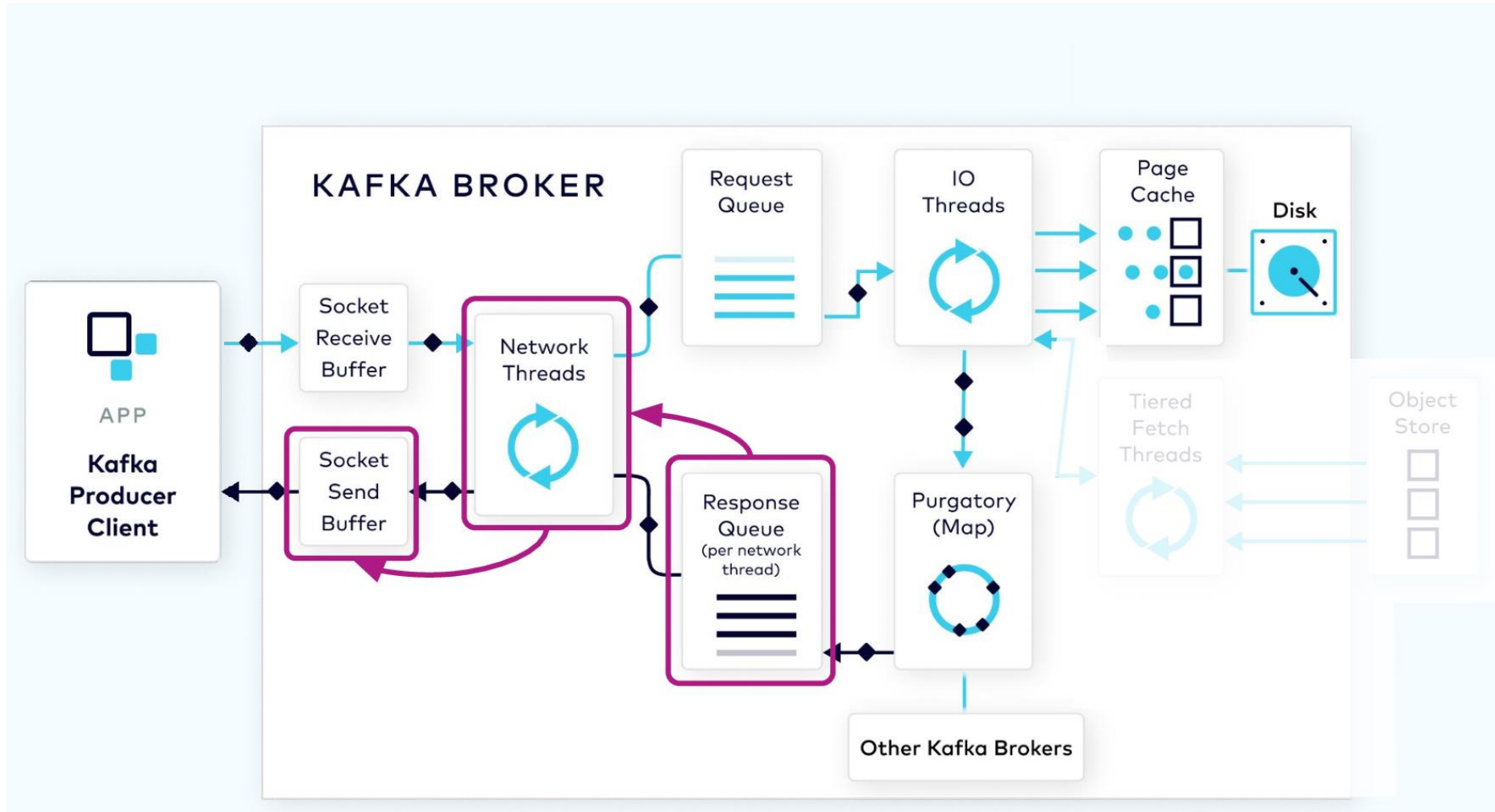
Kafka Manages Data and Metadata



In a Kafka cluster, the functions are divided into two planes: the **control plane** and the **data plane**. The control plane manages the cluster's metadata, while the data plane handles the actual data operations, such as reading from and writing to Kafka.

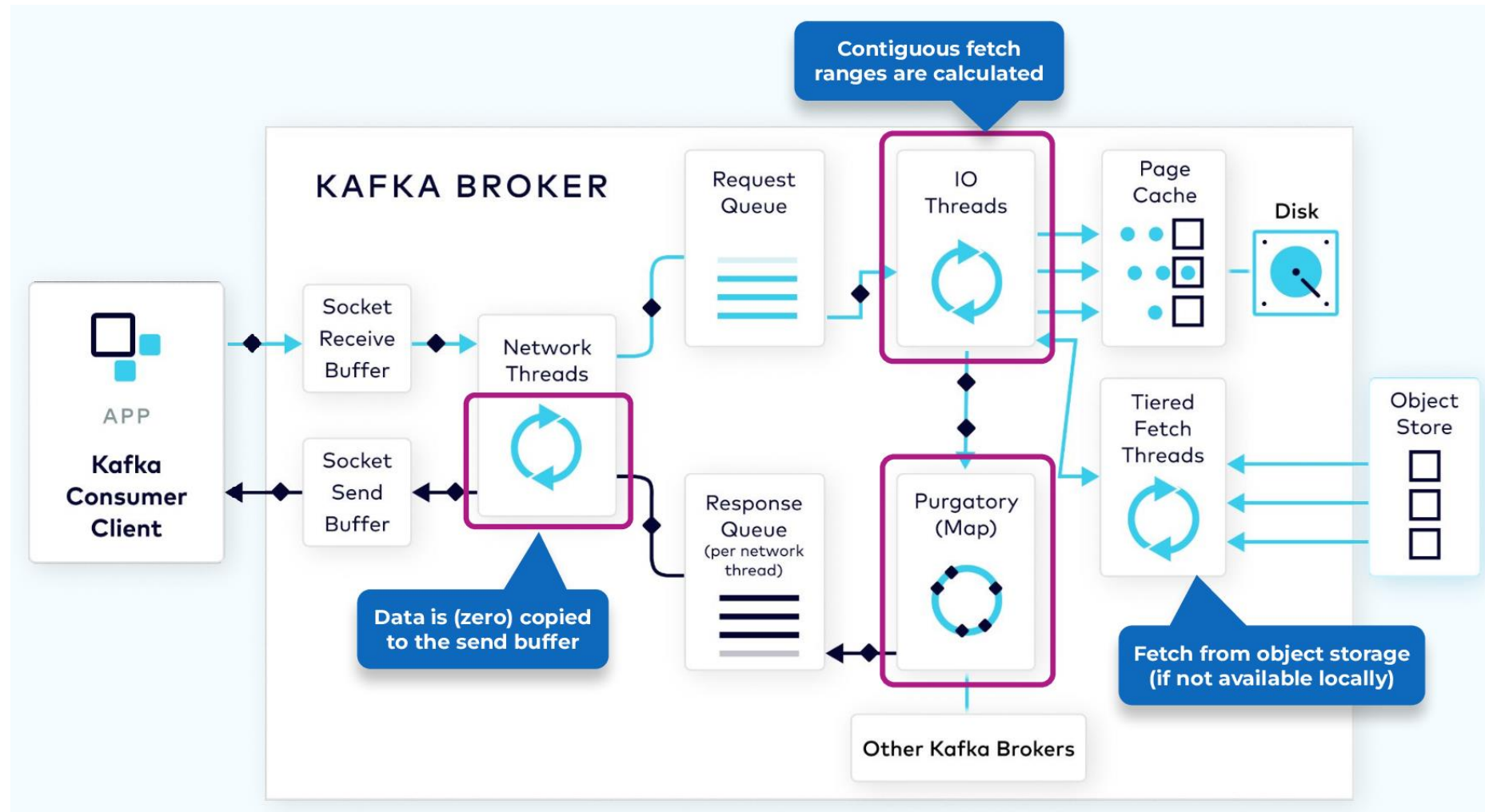
Inside the Apache Kafka Broker (Produce)

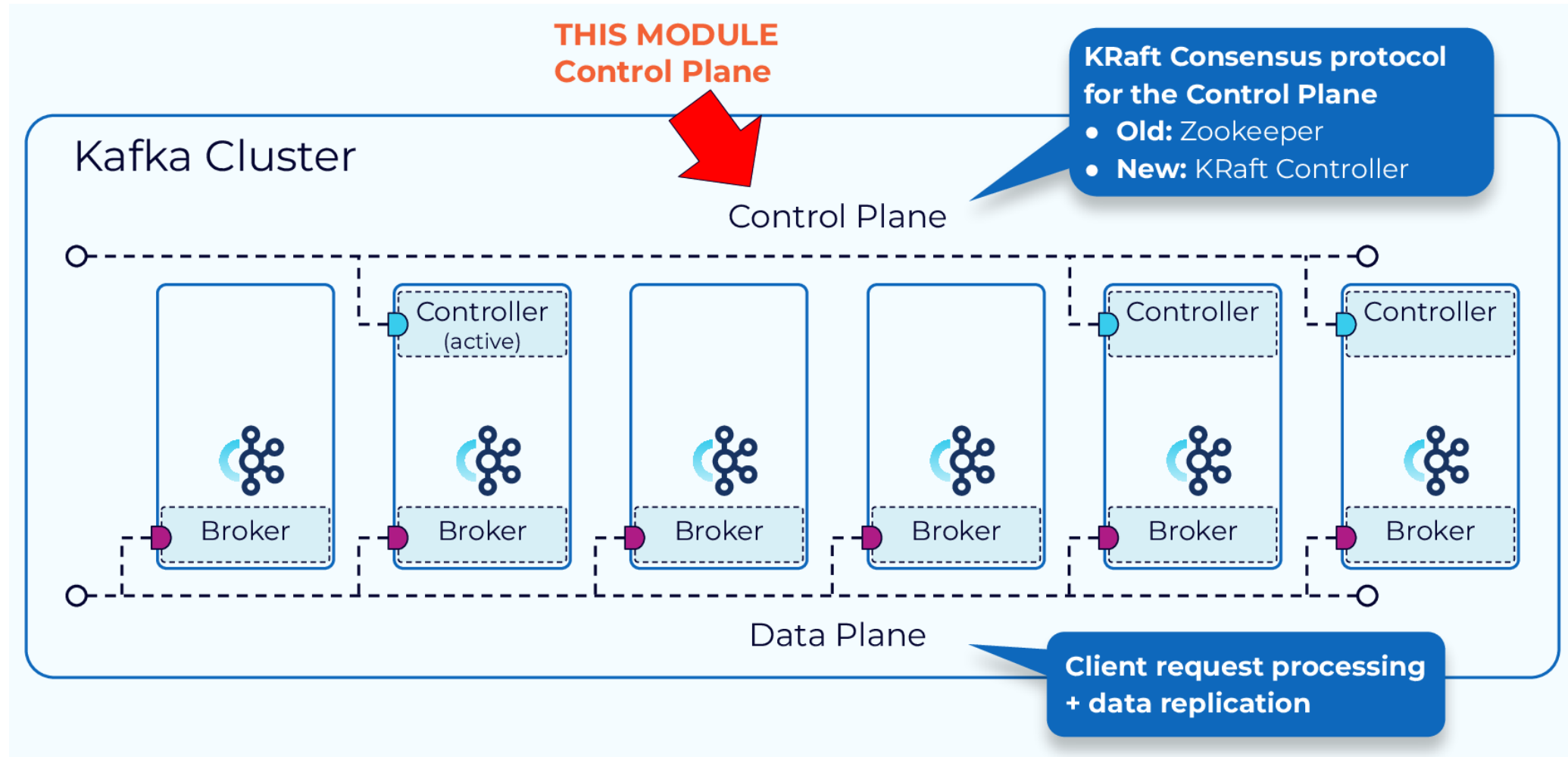
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Inside the Apache Kafka Broker (Consume)

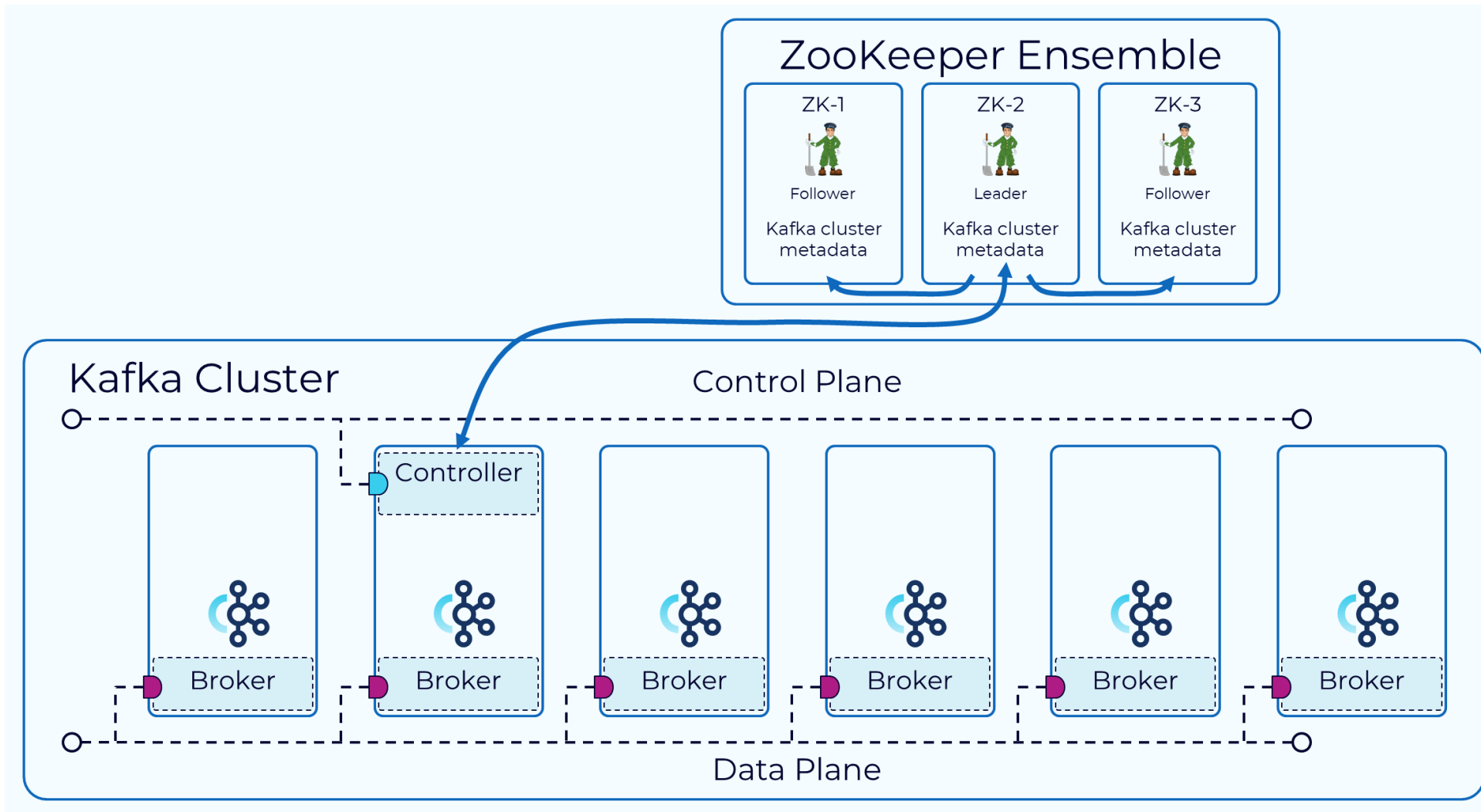
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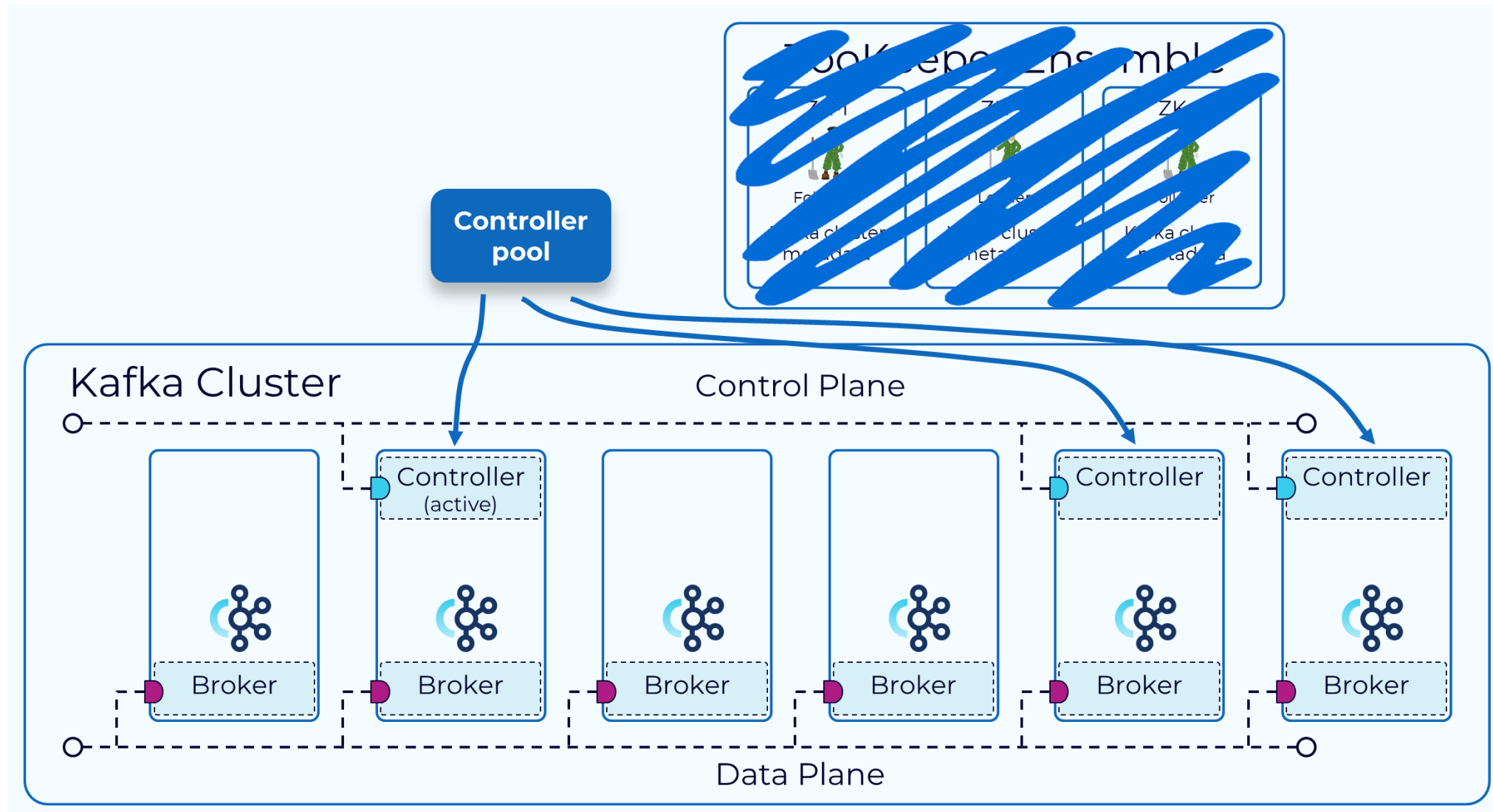




ZooKeeper Mode

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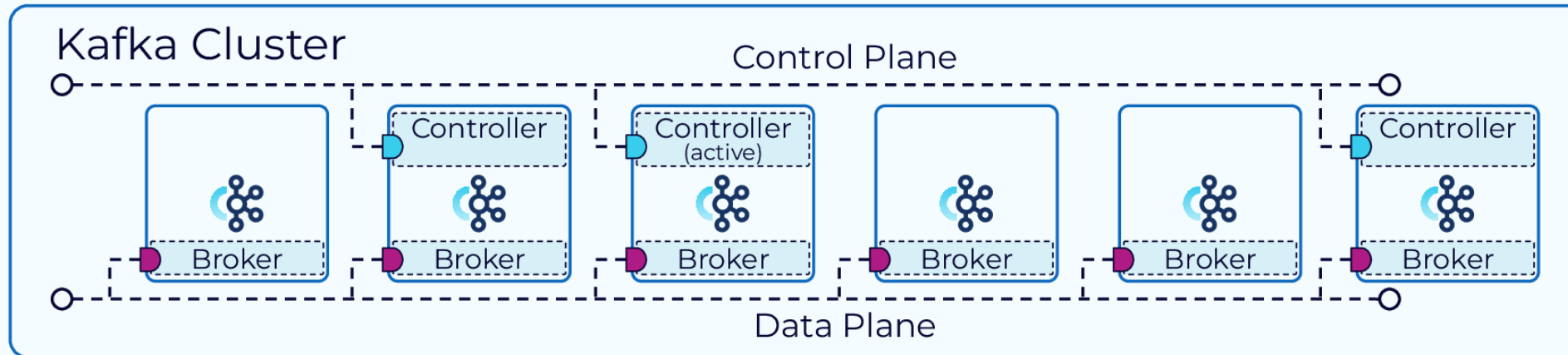
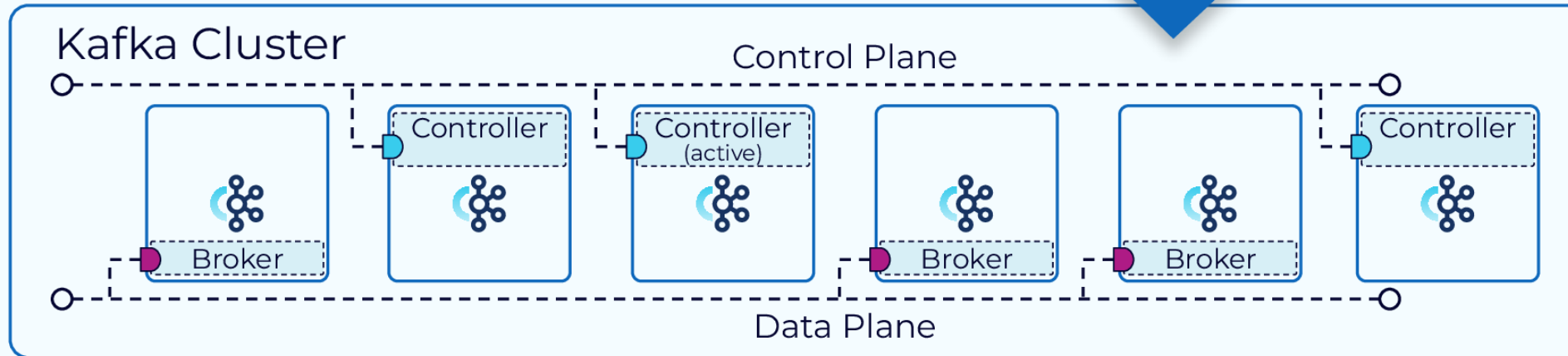


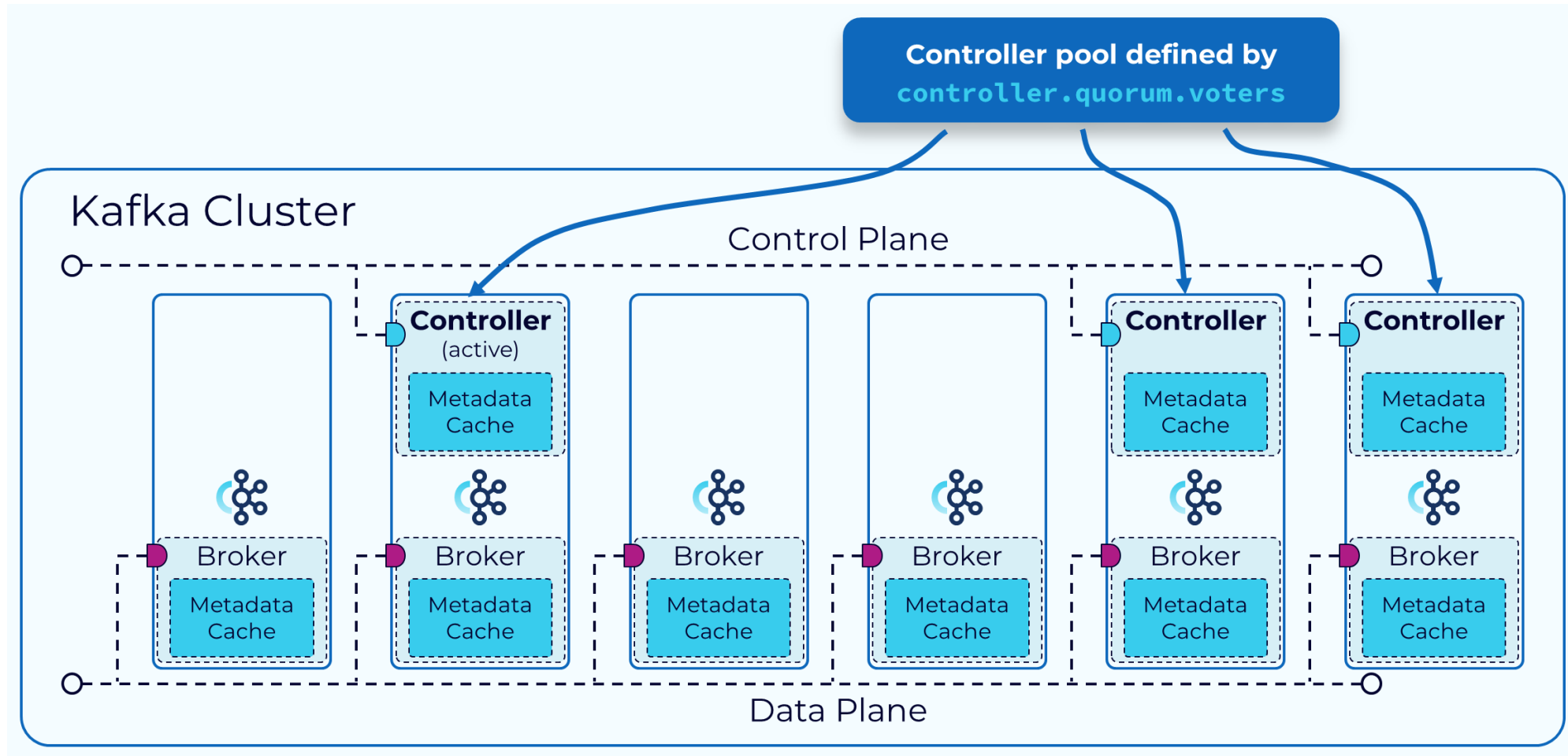
There are many advantages to the new KRaft mode, but we'll discuss a few of them here.

- **Simpler deployment and administration** – By having only a single application to install and manage, Kafka now has a much smaller operational footprint. This also makes it easier to take advantage of Kafka in smaller devices at the edge.
- **Improved scalability** – As shown in the diagram, recovery time is an order of magnitude faster with KRaft than with ZooKeeper. This allows us to efficiently scale to millions of partitions in a single cluster. With ZooKeeper the effective limit was in the tens of thousands.
- **More efficient metadata propagation** – Log-based, event-driven metadata propagation results in improved performance for many of Kafka's core functions.

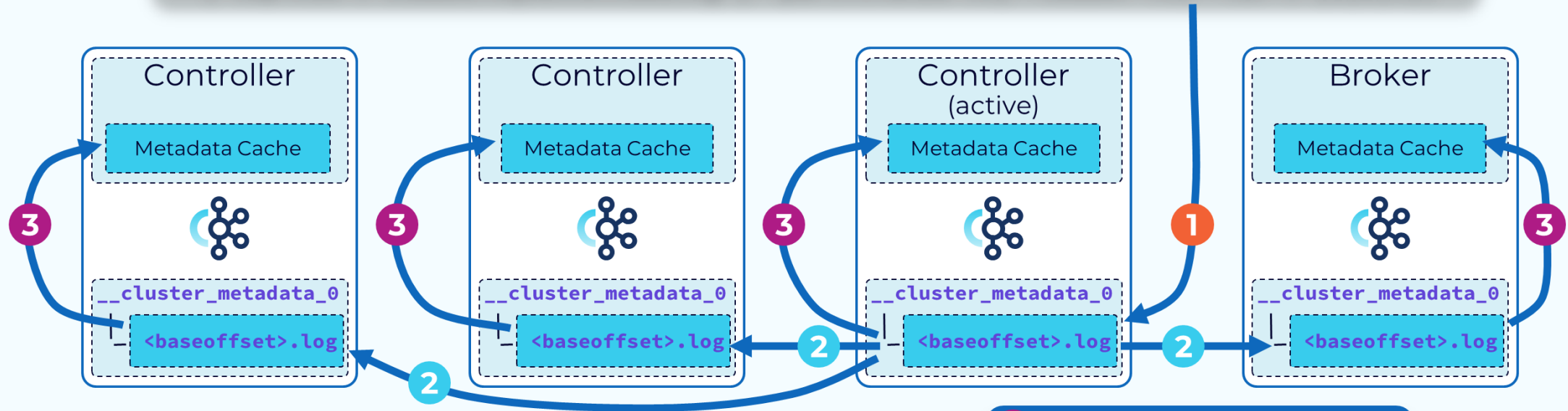
KRaft Cluster Node Roles

Cluster node role can be **controller**
or **broker** or **controller, broker**





1 Active controller writes the metadata change records to the `__cluster_metadata` topic, e.g.
`{"topicId":"JJmWDIT9Q9eXVDvCaBIJhg", "partitionId":0, "leader":1, "isr": [1,2,3]}`

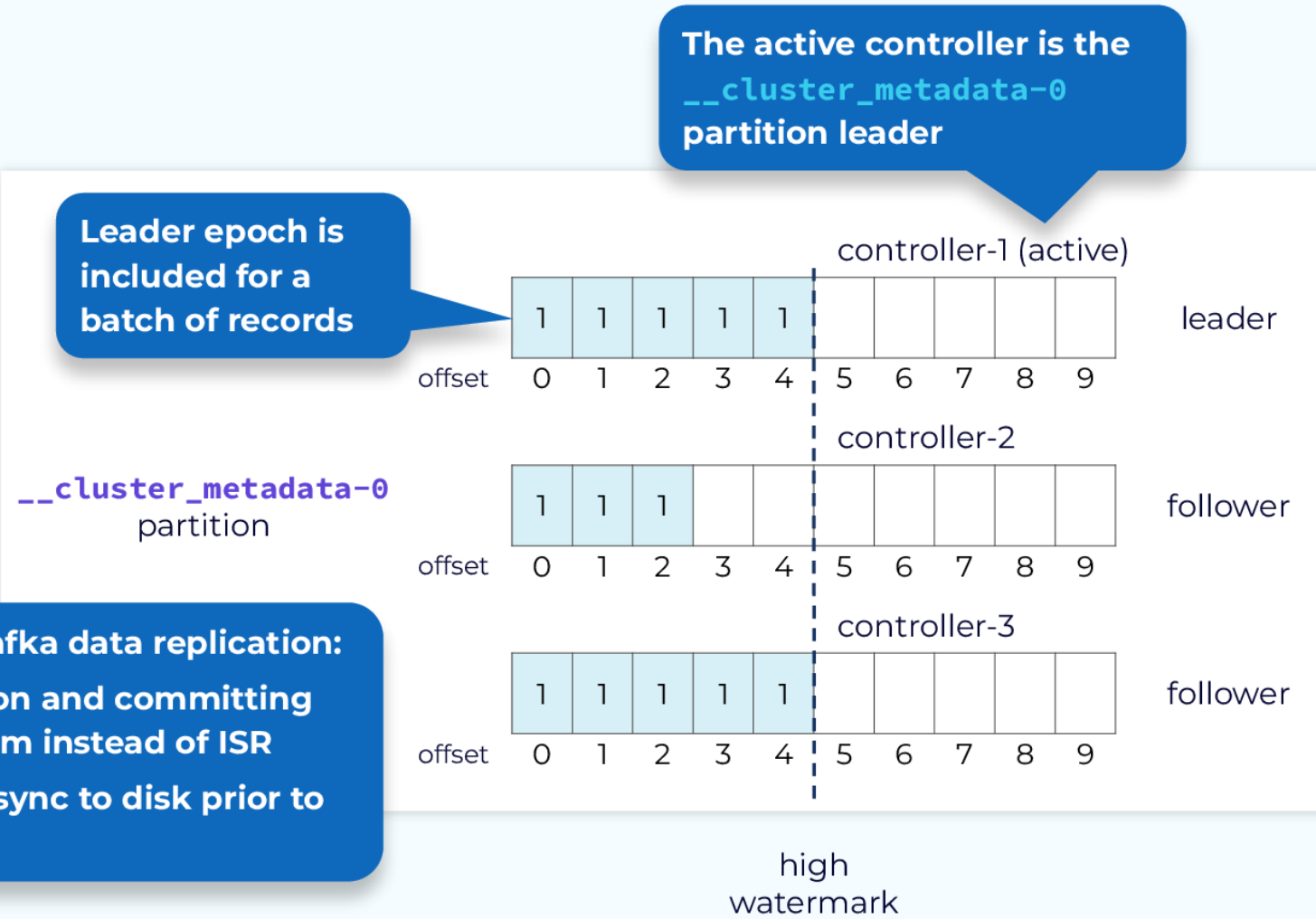


2 Controllers (followers) and brokers (observers) replicate `__cluster_metadata` topic events.

3 Controllers and brokers update metadata cache from committed log records

KRaft Metadata Replication

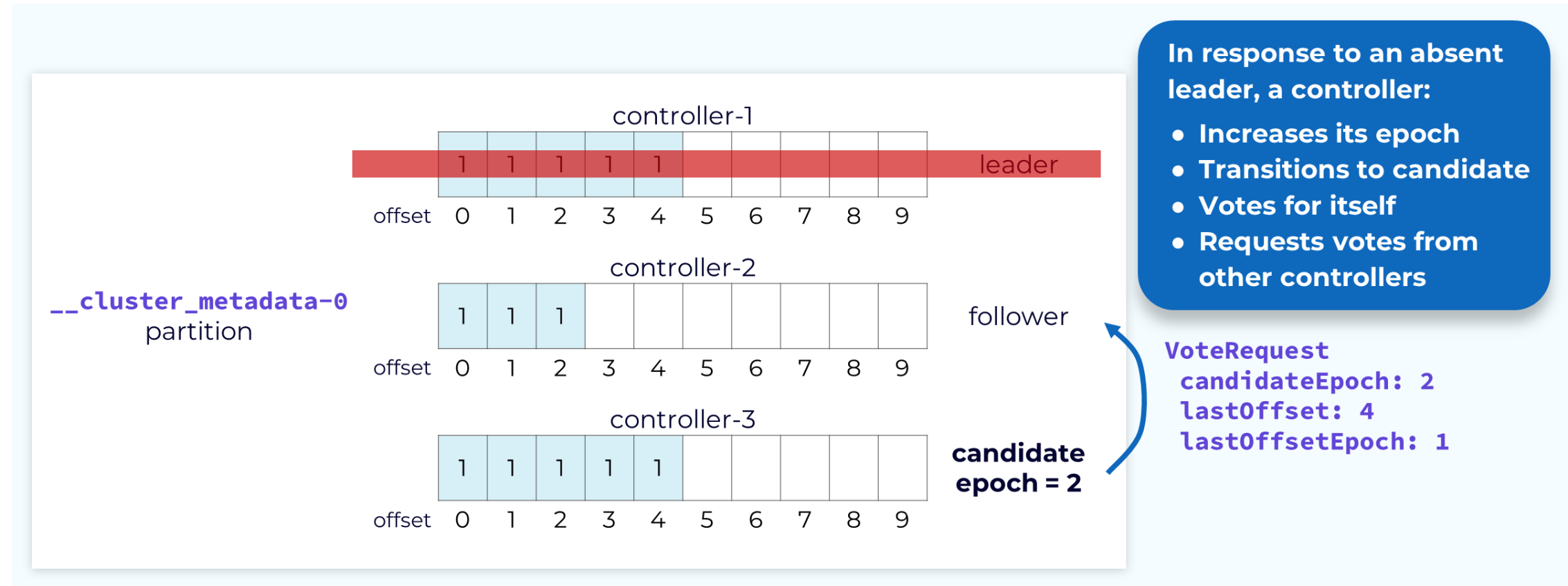
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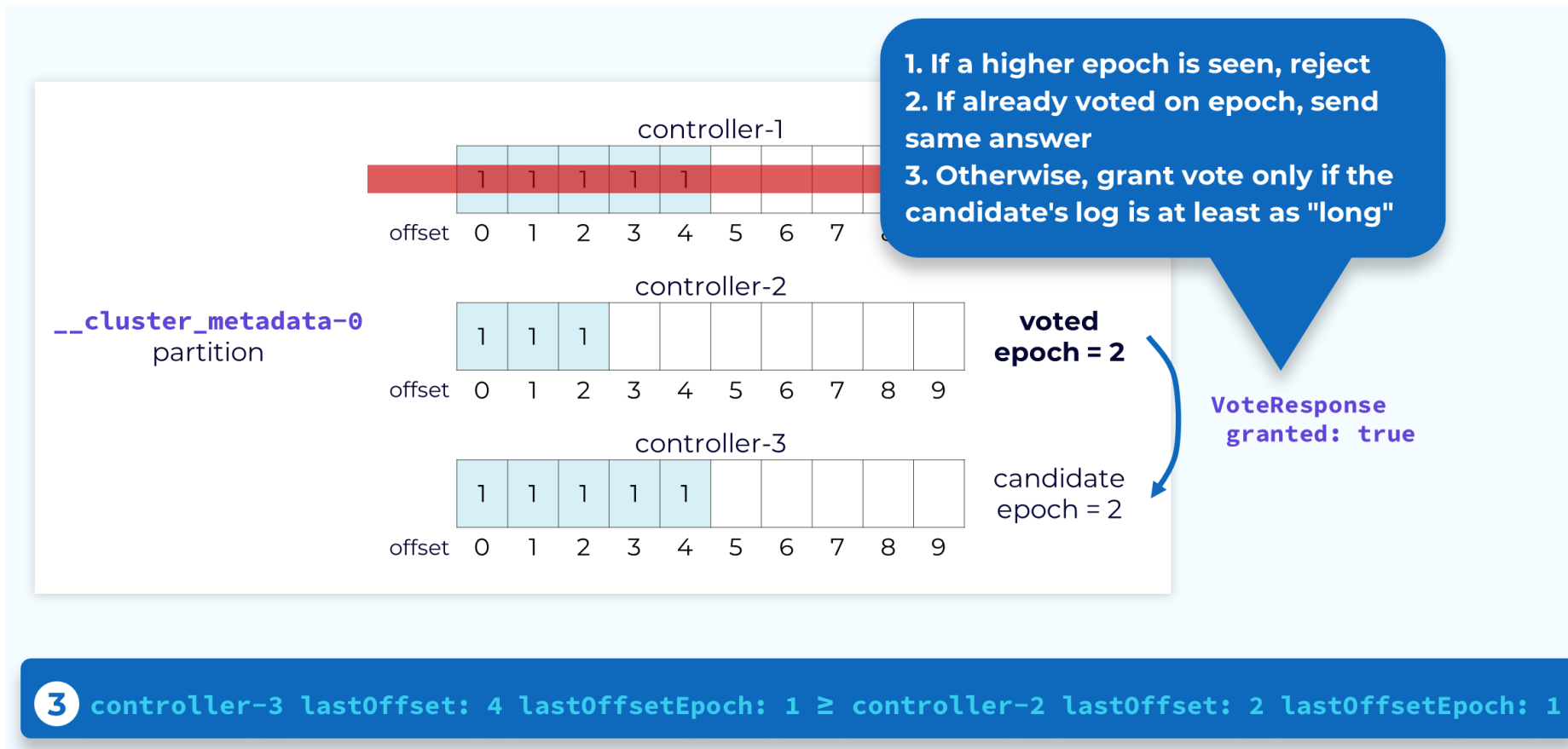
Difference to Kafka data replication:

- Leader election and committing record: quorum instead of ISR
- Records are fsync to disk prior to commit

Vote Request

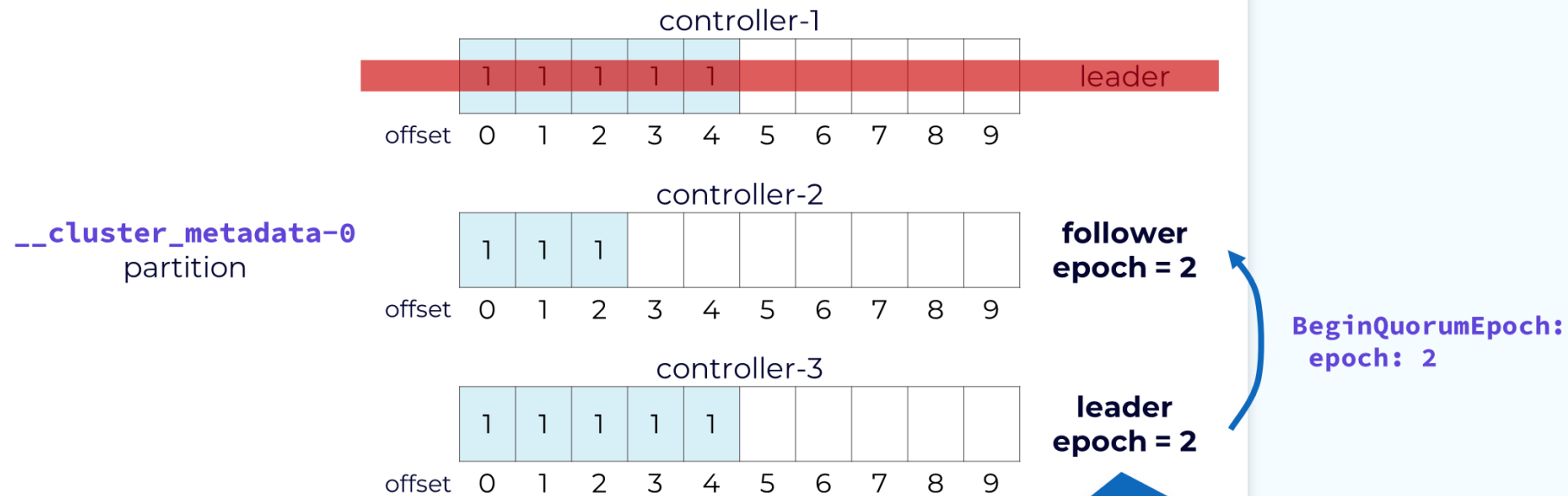


Vote Response



The first controller to achieve a majority of the votes becomes the new leader

Completion

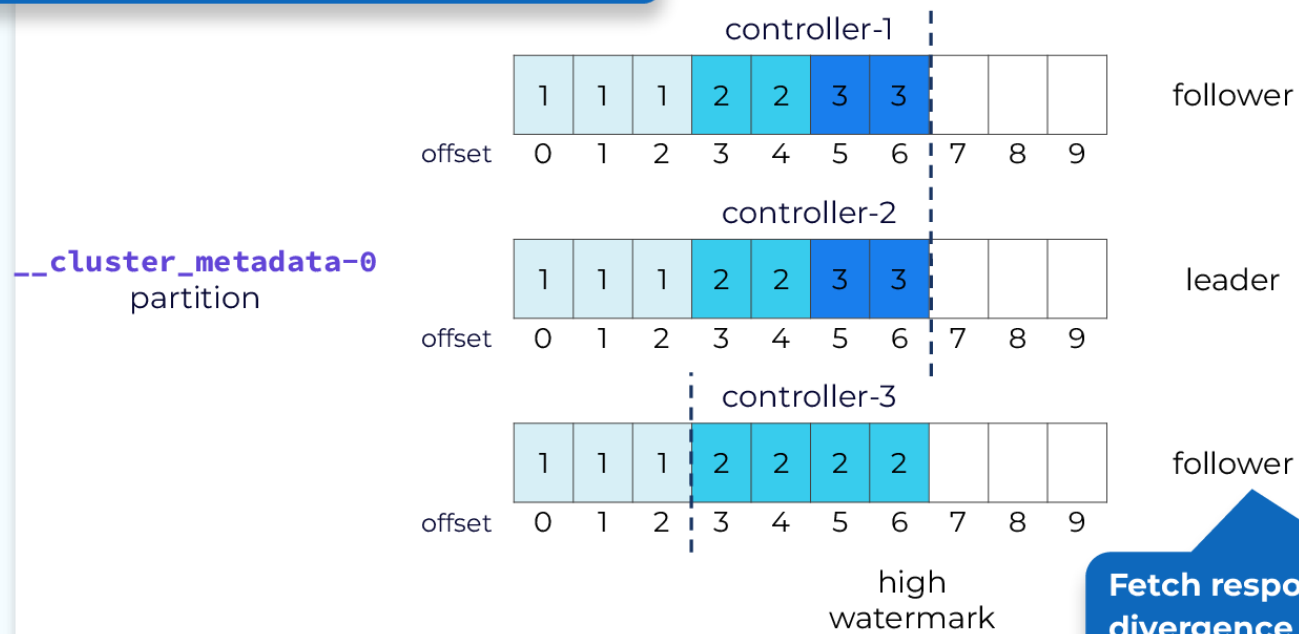


When a candidate receives the majority votes, it becomes the new leader and notifies other controllers the election is complete

Metadata Replica Reconciliation

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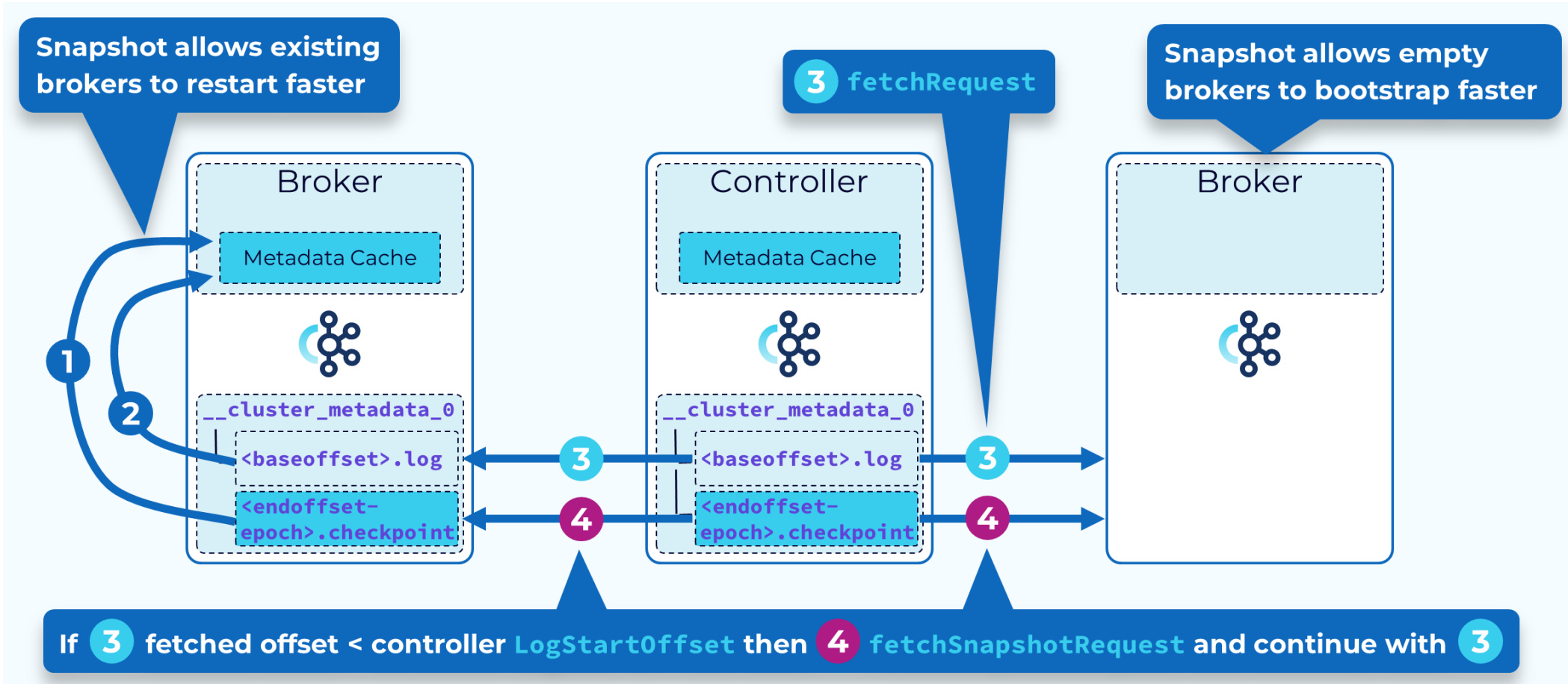
The same reconciliation process is used for both KRaft metadata and Kafka topic data



Fetch response will tell controller-3 divergence exists and it will truncate its replica to offset 5

When a Snapshot Is Read

Tos



Lab : Review Kraft Config 30 - Minutes